AWS Serverless Media Processing Architecture

Project Summary

This project implements a fully serverless media processing architecture using AWS services to handle image and video uploads, transformation, and secure global delivery.

Key AWS Services Used

- Amazon S3: File storage and Lambda trigger
- AWS Lambda: Serverless functions for media processing
- DynamoDB: Stores media metadata
- CloudFront: Content delivery network
- SQS: Message queue for processing jobs
- Elastic Transcoder: Converts media formats
- Route 53: DNS routing
- Cognito: Authentication & authorization
- IAM, Security Groups, CloudWatch: Security & Monitoring

Architecture Flow

1. User uploads media to S3
2. Lambda is triggered to process metadata
3. Tasks are queued in SQS
4. Worker nodes process and store back to S3
5. CloudFront delivers content globally
6. DynamoDB holds metadata; Cognito manages user access

Project Highlights

- Serverless & cost-efficient
- Secure & scalable
- Real-time processing & global delivery
- Integrated monitoring & role-based access control